

Andrew “Michael” Coggins

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EDUCATION

Texas A&M University College Station, Texas
Bachelor of Science in Electrical and Computer Engineering
minors in Mathematics and AI in Business

Graduation May 2028
Cumulative GPA: 3.5/4.00
University GPA: 3.4/4.00

TECHNICAL SKILLS

Relevant Coursework: Advanced Computer Architecture & Design, Digital System Design, Electrical Circuit Theory, Electronics & Magnetism, Discrete Structures, Program Design & Concepts, Calculus I-III

Tools: Vivado, LTspice, IVerilog, GTKWave, Google Cloud, REST APIs, Git, Claude Code, VS Code, GitHub

PROJECTS

Single Cycle ARMv8 Processor | Verilog HDL, Computer Architecture, IVerilog, ARMv8 Assembly

- Designed and verified a **single-cycle ARMv8 RISC processor** in **Verilog HDL**, implementing a 64-bit ALU, multi-format Sign Extender, control unit, and branch-resolving NextPC logic.
- Validated instruction execution and data path integrity through comprehensive simulation testbenches.

Interactive Pi 5 Hardware Visualizer | Three.js, WebGL, Hardware Architecture, HTML/CSS, Claude Code

- Engineered an in-browser 3D scale model mapping 23 Raspberry Pi 5 board components
- Built dynamic click-to-inspect ray casting panels documenting component functions, electrical specs, and signal paths to visualize hardware architecture.

Class Schedule Optimizer | Python, Flask, REST APIs, Web Scraping, Claude Code

- Built a data pipeline ingesting **21.6k+ course sections**, combining RateMyProfessors and grade archives (Anex.us) via fuzzy string matching.
- Developed weighted scoring and cart-syncing logic to evaluate GPA distributions and push to schedules directly

Autonomous Web Studio Agent | Claude Code, Vercel, Git

- Authored custom Claude Code agent skills to orchestrate multi-step autonomous workflows, automating client research, prompt-driven content synthesis, and site layout generation.
- Architected an agentic build pipeline mapping structured AI outputs into templates for automated Vercel deployment

LEADERSHIP & INVOLVEMENT

One Army, Texas Aggie Men United | Class Candidate Advisor

September 2025 – Present

- One of eight leaders in charge of onboarding and mentoring new members for an organization of 90+ members
- Demonstrated leadership and project management skills within team as we marketed and executed part of the annual Gladiator Dash, a major fundraising event. In 2026, we raised a record **\$185,000**, a **2.75% increase** from last year.

Institute of Electronics & Electrical Engineers (IEEE) | member

September 2025 – Present

- Networked with industry professionals and fellow electrical engineers while increasing practical experience with hardware components such as **Breadboards, FPGAs** and **Raspberry Pi**. Tools used **MultiSim, Vivado, IVerilog, GT-Wave**.
- Constructed multi-stage sequential logic state machines (combination locks) using discrete logic gates/flip-flops and designed Sallen-Key op-amp active filters, verifying frequency response and signal integrity.

Texas A&M Baja SAE | Business Lead

August 2026 – Present

- Directed **cross-functional** recruitment and **operational management** across accounting, sponsor relations, social media, financing, and operations sub teams while serving as primary liaison to Project Managers and university sponsors.
- Managed corporate partner relations, spearheading sponsorship acquisition, pitch execution, and industry stakeholder engagement to secure critical funding and material donations.

EMPLOYMENT AND INTERNATIONAL EXPERIENCE

Texas A&M Study Abroad | Queens University, Belfast, Northern Ireland, UKGB

Summer 2026

- Toured AMD Headquarters in Dublin. Met with R&D leaders about AMD's NPU/CPU/GPU compute stack, AI for efficiency benchmarking processes, and Xilinx FINN's FPGA Accelerated Quantized Neural Network Interface

